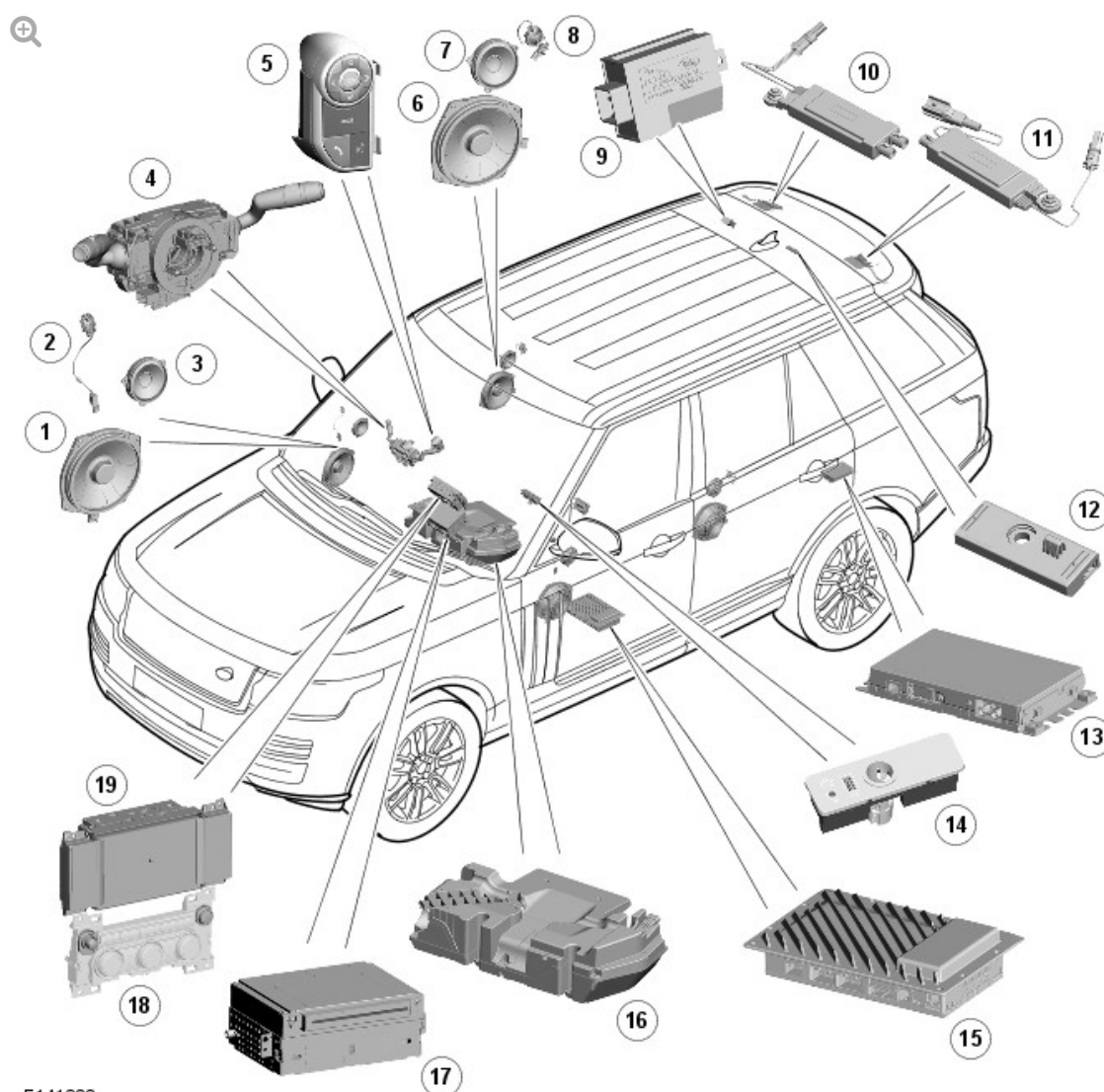


2016.0 RANGE ROVER (LG), 415-01

INFORMATION AND ENTERTAINMENT SYSTEM

DESCRIPTION AND OPERATION

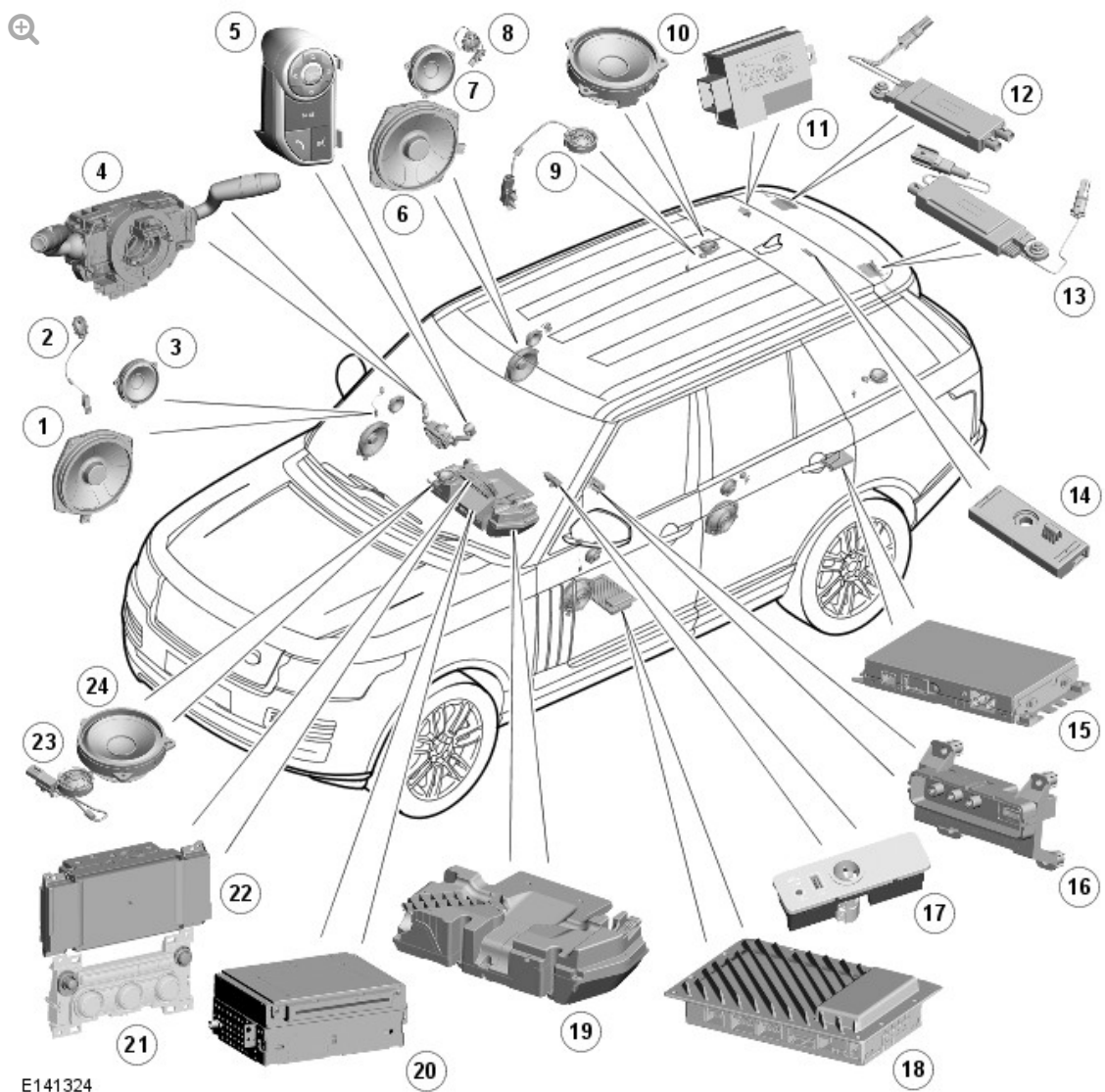
COMPONENT LOCATION - MERIDIAN SYSTEM



E141323

ITEM	DESCRIPTION
1	Front bass speaker (2 off)
2	Front mid-range speaker (2 off)
3	Front tweeter speaker (2 off)
4	Clockspring
5	Left steering wheel switchpack
6	Rear bass speaker (2 off)
7	Rear mid-range speaker (2 off)
8	Rear tweeter speaker (2 off)
9	RF filter - Tailgate wiper motor
10	Antenna amplifier - FM2/Digital Radio/TMC/(VICS Japan only)
11	Antenna amplifier - AM/FM
12	RF filter - High mounted stop lamp
13	Digital Radio control module or satellite radio control module (NAS only)
14	Portable audio interface panel
15	Audio amplifier module
16	Sub-woofer
17	Integrated Audio Module (IAM)
18	Integrated Control Panel (ICP)
19	Touch Screen (TS)

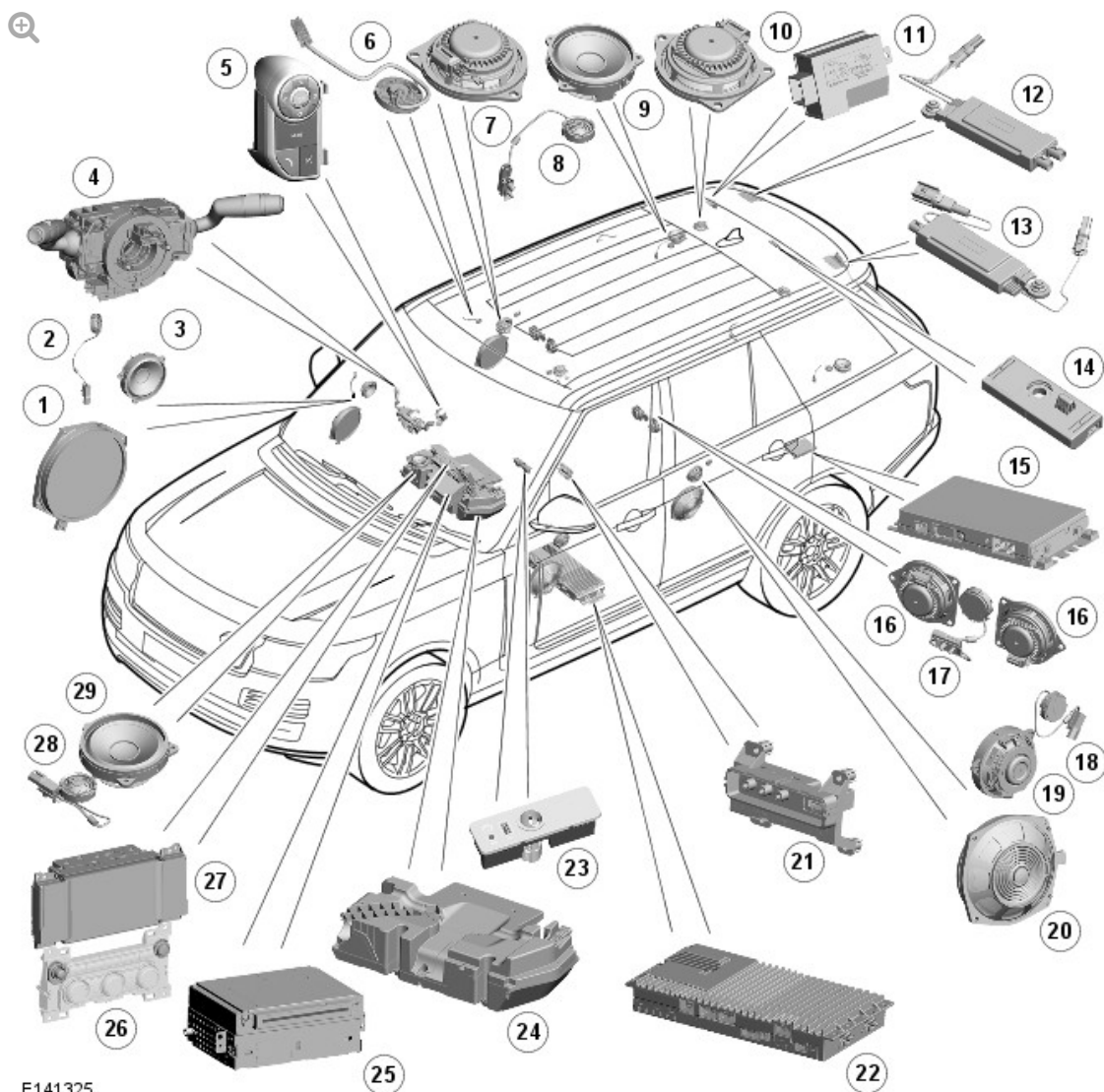
COMPONENT LOCATION - MERIDIAN SURROUND SYSTEM



ITEM	DESCRIPTION
1	Front bass speaker (2 off)
2	Front tweeter speaker (2 off)
3	Front mid-range speaker (2 off)
4	Clockspring
5	Left steering wheel switchpack
6	Rear bass speaker (2 off)
7	Rear mid range speaker (2 off)
8	Rear tweeter speaker (2 off)
9	Rear surround tweeter speaker (2 off)
10	Rear surround mid-range speaker (2 off)

11	RF filter - Tailgate wiper motor
12	Antenna amplifier - FM2/Digital Radio/TMC/(VICS Japan only)
13	Antenna amplifier - AM/FM
14	RF filter - High mounted stop lamp
15	Digital Radio control module or satellite radio control module (NAS only)
16	Audio Video Input/Output (AVIO) panel
17	Portable audio interface panel
18	Audio amplifier module
19	Sub-woofer
20	Integrated Audio Module (IAM)
21	Integrated Control Panel (ICP)
22	Touch Screen (TS)
23	Instrument panel surround tweeter speaker
24	Instrument panel surround mid-range speaker

COMPONENT LOCATION - MERIDIAN SIGNATURE REFERENCE SYSTEM



ITEM	DESCRIPTION
1	Front bass speaker (2 off)
2	Front tweeter speaker (2 off)
3	Front mid-range speaker (2 off)
4	Clockspring
5	Left steering wheel switchpack
6	Microphone - Dynamic Equalization (4 off)
7	Front headliner surround low profile mid-range speaker (2 off)
8	Luggage compartment tweeter speaker (2 off)
9	Luggage compartment surround mid-range speaker (2 off)
10	Rear headliner surround mid-range speaker (2 off)

11	RF filter - Tailgate wiper motor
12	Antenna amplifier - FM2/Digital Radio/TMC/(VICS Japan only)
13	Antenna amplifier - AM/FM
14	RF filter - High mounted stop lamp
15	Digital Radio control module or satellite radio control module (NAS only)
16	Seat back surround low profile mid-range speaker (4off)
17	Seat back tweeter speaker (2 off)
18	Rear tweeter speaker (2 off)
19	Rear mid-range speaker (2 off)
20	Rear bass speaker (2 off)
21	Audio Video Input/Output (AVIO) panel
22	Audio amplifier module
23	Portable audio interface panel
24	Sub-woofer
25	Integrated Audio Module (IAM)
26	Integrated Control Panel (ICP)
27	Touch Screen (TS)
28	Instrument panel surround tweeter speaker
29	Instrument panel surround mid-range speaker

OVERVIEW

The audio system is available in three versions:

- Meridian System 384W - 13 speakers including dual voice coil subwoofer and single slot CD (compact disc) player
- Meridian Surround System 880W - Meridian Trifield, Dolby Pro Logic IIx and DTS Neo:6 Surround Sound System, 19 speakers including dual voice coil subwoofer and single slot CD player

- Meridian Signature Reference System 1700W Meridian Trifield, Meridian Trifield 3D, Dolby Pro Logic IIx, Dolby Pro Logic IIz, and DTS Neo:X Surround Sound System - 29 Speakers including dual voice coil subwoofer and single slot CD player.

All sound systems have separate amplifiers and include:

- Auxiliary input
- MP3-compatible, single slot CD
- AM / FM radio
- HD Radio™
- Digital Radio Control Module (Europe and Asia only) or Satellite Radio control module (NAS only).
- EON, RDS, PTY, TA
- Bluetooth® and audio streaming
- 40GB Hard Disc Drive (HDD) for navigation and virtual CD player
- Front media interface.

All sound systems are specified with:

- Hard Drive Navigation system (except Japan and Asian markets) with virtual CD player and single slot DVD (digital versatile disc).
- Interactive 'Say What You See' (SWYS) Advanced Voice system.

A range of audio systems are available combining radio and CD / DVD functions, satellite radio or Digital Radio, MP3 disc compatibility, RDS, PTY and TA functions. The Integrated Audio Module (IAM) is a combined radio tuner, CD / DVD player and incorporates a 40GB hard drive for use by the audio for storing music and also stores the operating system and files for the navigation system (not applicable to Asian market vehicles).

A portable audio interface panel allows for the connection of a range of portable audio devices to the car's audio system. The panel is controlled

through the Integrated Audio Module (IAM), with play back through the vehicle's speaker system. This system allows the user to import data from their personal portable media player to interface with the vehicle, including iPod® and other MP3 players, or USB (universal serial bus) mass storage devices such as memory sticks. MP3 players can also be controlled through the Touch Screen (TS) if they are configured as mass storage devices. The portable audio interface panel includes a 3.5mm auxiliary jack-plug socket, plus a USB2 connector which allows connectivity for a wide variety of USB devices, including iPods via the USB. The USB port also includes a charging function, although it does not support a USB hub. The maximum charging current supplied is 500ma. The user can connect either an iPod or a USB device to the USB connector, and select the appropriate function from the TS.

Digital Radio (DAB, DAB+ and DMB) is available Europe, Canada, Australia and Asian markets and gives access to digital radio channels for better sound quality and enhanced functionality depending on local service availability. A Digital Radio control module processes the received digital signals. The system receives reception signals from the following sources to ensure optimum signal strength:

- Digital Radio band L antenna located in the roof pod antenna module
- Digital Radio band III antenna located in the rear spoiler.

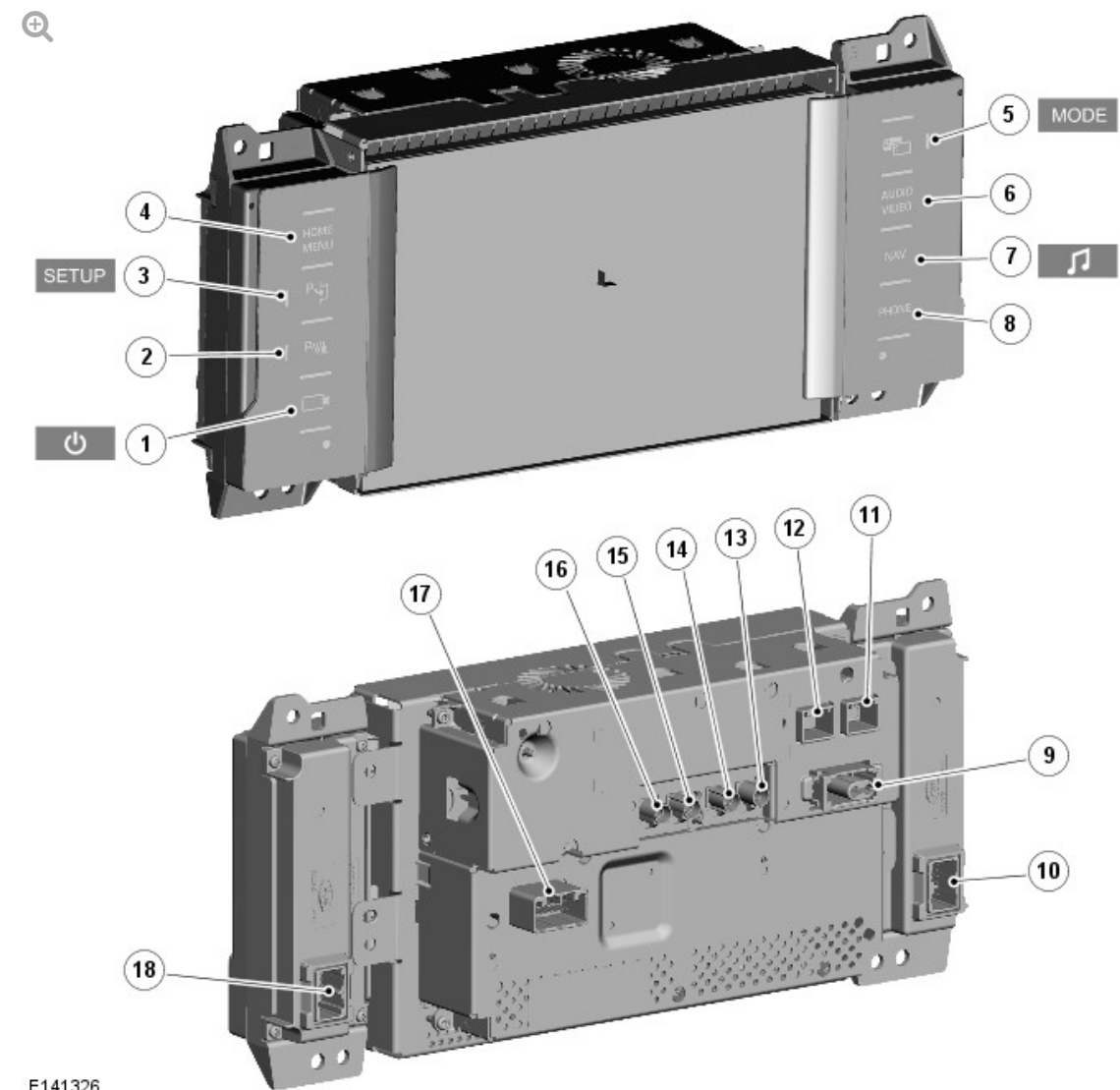
For NAS vehicles, the digital format adopted is Satellite Digital Audio Radio Service (SDARS) which specifically links to the Sirius® network. The subscription system operates in the S-band frequency range, and as a result of the use of satellite transmission, has the ability to provide CD quality audio broadcasts over very large areas (typically continents). The satellite radio control module processes digital signals from the satellite radio antenna located in the roof pod.

Primary user control of the audio system is via the TS and the Integrated Control Panel (ICP), which are located in the center of the instrument panel. Control signals from the ICP are relayed on the MS (medium speed) CAN

(controller area network) comfort bus to the IAM. The IAM relays the control signals to the rest of the audio system on a Media Oriented System Transport (MOST) ring. The TS is the timing master for the MOST ring and also hosts a gateway function between the MS CAN comfort bus and the MOST ring.

DESCRIPTION

TOUCH SCREEN (TS)



E141326

ITEM	DESCRIPTION
1	Camera view on/off or screen on/off soft key
2	Park distance control on/off soft key

3	Park assist on/off or Settings menu soft key
4	Home menu button
5	Dual view on/off or audio/video Mode soft key
6	Audio/Video menu soft key
7	Navigation menu or audio settings soft key
8	Telephone menu soft key
9	Connector - MOST
10	Left TS switchpack connector
11	Connector - LVDS output to instrument cluster
12	Connector - LVDS input from Integrated Audio Module (IAM) or Navigation control module (Japan or Asia)
13	Connector - CVBS video input from parking aid camera control module or rear view camera
14	Connector - CVBS video input from TV control module or RSE control module
15	Connector - CVBS video input from IAM
16	Connector - CVBS video output to RSE control module
17	Connector - Power, ground, Medium Speed CAN bus, steering wheel switch input
18	Right TS switchpack connector

The Touch Screen (TS) is located in the center of the instrument panel. The TS comprises an 8 inch color, touch sensitive display, with two separate touch sensitive TS switchpacks located on either side.

Two versions of the TS are available; single view and dual view. Additionally TS switchpacks differ depending on system specification; for example, if Park Assist, navigation, dual view or rear view camera is specified on the vehicle, these soft keys will replace Setup, Audio Settings, Mode and display on/off respectively. The functions can still be accessed via the home menu using the screen on the TS.

The TS is the Bus Master for the Media Oriented System Transport (MOST) system and contains the timing master for the MOST system.

When the vehicle systems become active the TS is woken up by CAN bus activity and subsequently wakes up the other audio modules via the MOST.

The TS is the primary user interface of the audio system. It communicates with the other components of the audio/infotainment system on the MOST ring and allows control of the audio system and other infotainment systems from a single point. No configuration procedure is required if the TS or TS switchpacks are replaced.

Calibration of the TS using approved Land Rover diagnostic equipment enables updates to be downloaded as new technology becomes available or any fault concerns require software updates.

The rear of the TS has electrical connectors for power, ground, MS CAN comfort bus, Low Voltage Differential Signalling (LVDS) and Composite Video Baseband Signal (CVBS) video and MOST. The usage of the LVDS and CVBS connectors differs depending on the equipment level of the vehicle.

The TS switchpacks comprise a number of touch sensitive soft keys. The soft keys are not visible when the system is powered down. When the system is active the soft keys are illuminated. When a soft key is pressed, a line above and below the soft key is illuminated to indicate the press has been accepted.

Each TS switchpack has a single connector which provides electrical connections to the control modules for the soft key functions and tell-tale LED (light emitting diode) illumination. Additionally, each switchpack is connected to other vehicle systems via the MS CAN comfort bus.

DUAL-VIEW TOUCH SCREEN (where fitted)



E141327

ITEM	DESCRIPTION
1	Dual-view softkey

The dual-view TS enables the passenger and driver to view completely different images from their respective seating positions. This technology has provided a solution for the legal issues attached to viewing moving images whilst the vehicle is in motion. It is not possible for the driver to view moving images with an active speed signal, but the passenger can view DVD and TV independent of the driver.

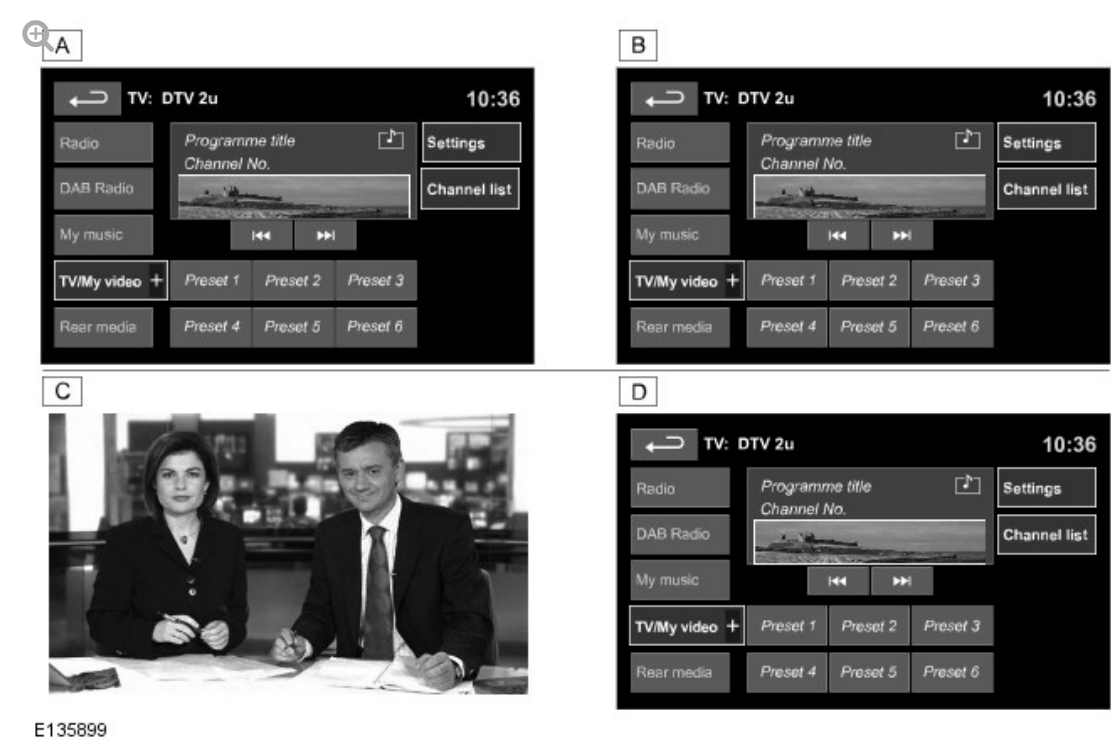
Vehicles fitted with a dual view TS are supplied with a headphone transmitter located in the headliner and one pair of Whitefire™ infra-red wireless headphones. This allows the front seat passenger to view and listen to music, DVD and TV without disturbing the driver.

NOTE:

Due to legislation the NAS markets do not receive this option. A single view display is available in these markets.

The dual-view TS uses Parallax Barrier Shutter Technology to alternately hide and reveal columns of pixels to the left and right hand views of the screen. The display comes with a specially designed agar coating to help prevent sunlight bleaching.

To access a TV or video image when the vehicle is in motion and single view is selected, the dual view soft key on the TS switchpack should be pressed by either the driver or the passenger. This will then switch the TS to dual-view mode allowing the passenger to view TV or video, but not the driver. A second press of the soft key will change the TS back to single view.



ITEM	DESCRIPTION
A	Passenger view before dual-view button pressed
B	Driver's view before dual-view button pressed

C	Passenger view after dual-view button pressed
D	Driver's view after dual-view button pressed

Once dual-view has been selected, the driver can change the current screen without affecting the passengers view by the source on the TS.

The audio system can only broadcast one audio source. Therefore, the TV / video source that is current for the passenger will also be the audio the driver can hear. The passenger's can choose to use headphones to listen to the sound source accompanying the TV / video. This allows the driver to listen to a different audio source or navigation commands via the vehicle speaker system.

The driver's view is also event driven, for example, if reverse gear were to be selected, the rear view camera will be displayed automatically, overriding the currently displayed information.

CLOCK

The CJB (central junction box) contains the master clock functionality. Other vehicle infotainment system modules that require clock functionality use the time supplied from the CJB.

The clock is available to any control module that is connected to an interconnecting bus, for example, either of the CAN buses or the MOST ring.

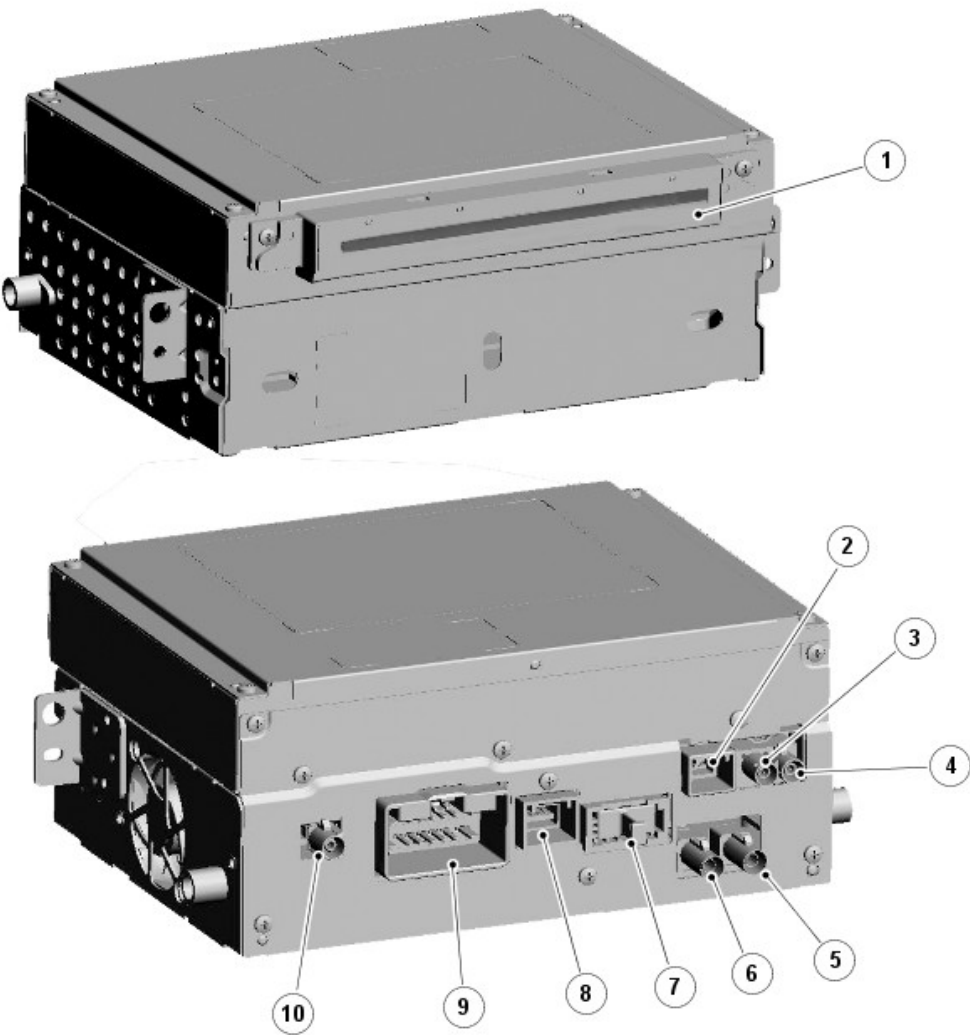
The clock display is configurable to show in AM / PM or 24 hour format. Midnight is shown as 12:00AM or 00:00 respectively. The default condition, if not specified, after power on or delivery, should default to 12:00PM or 00:00. Depending upon the market set, the clock will default to either 12 or 24 hour format.

The time is adjusted from the TS. Under conditions when any bus system could be asleep or shut down, the TS does not allow clock adjustments.

INTEGRATED AUDIO MODULE (IAM)

NOTE:

IAM3 version shown, not all connectors are present on IAM versions 1 and 2



E141328

ITEM	DESCRIPTION
1	CD/DVD loading slot
2	Connector - LVDS output to TS (IAM2 and 3 versions only)
3	Connector - AM/FM antenna amplifier input (All IAM versions)
4	Connector - FM2/TMC/(VICS Japan only) antenna amplifier input (All IAM versions)
5	Connector - Global Positioning System (GPS) antenna input (IAM3 version only)

6	Connector - VICS beacon antenna input (Japan only) (IAM3 versions only)
7	Connector - MOST input/output (All IAM versions)
8	Connector - Universal Serial Bus (USB) input from portable audio interface panel (All IAM versions)
9	Connector - Power, ground, AM/FM antenna amplifier power output, microphone inputs, auxiliary inputs from portable audio interface panel (All IAM versions)
10	Connector - CVBS video output to TS

The Integrated Audio Module (IAM) is located in the instrument panel, behind the Integrated Control Panel (ICP).

Three levels of IAM are available; IAM 1, IAM 2 and IAM3. The following table shows the features supported by each IAM version.

FEATURE	IAM1	IAM2	IAM3
CD/DVD	Single CD	Single CD/DVD	Single CD/DVD
Bluetooth®	Yes	Yes	Yes
USB	Yes	Yes	Yes
Virtual CD (Hard Disc Drive (HDD))	No	No	Yes
Compact Disc DataBase (CDDb)	No	No	Yes
Navigation (40GB HDD)	No	No	Yes
Voice control	No	No	Voice (hi-line and memo)
Microphone	Yes	Yes	Yes
Audio Auxiliary input	Yes	Yes	Yes
Amplifier	External	External	External
Radio Tuners	2	2	2
	USA: dual tuner with HD Radio™	USA: dual tuner with HD Radio™	USA: dual tuner with HD Radio™ and RDS TMC
	Europe: dual tuner	Europe: dual tuner	Europe: dual tuner with RDS TMC

			Japan: dual tuner with FM VICS
HD Radio™ (USA Only)	Yes	Yes	Yes
RDS (Europe) / RBDS (NAS)	Yes/Yes	Yes/Yes	Yes/Yes
MOST for Navigation	No	No	Yes
Video output	No	Yes (one source)	Yes (one source)

The IAM3 has the following features, in addition to those supported by IAM1 and IAM2:

- Navigation 40GB Hard Disc Drive (HDD) (30GB used for Navigation)
- TMC/(VICS Japan only)
- Off Road Navigation
- 40GB HDD (10GB used for music storage) - virtual CD/DVD - allows the storage of up to 10 CDs to create a 'virtual' multichanger.
- Voice recognition
- DVD output to Rear Seat Entertainment (RSE) control module
- Wireless headphones.

All versions of the IAM contain the following functionality:

- AM/FM radio tuner with diversity
- HD Radio™ receiver (NAS only)
- CD/MP3 player (Single CD)
- Auxiliary audio/video is available via a portable audio interface panel. The panel allows for the connection of portable audio/video via 2 USB's, a 3.5mm jack plug and iPod connectivity.
- Bluetooth® and audio streaming.

When the vehicle systems become active, the TS is woken up by CAN bus activity and subsequently wakes up the IAM via the MOST.

The IAM incorporates an AM (amplitude modulation)/FM (frequency modulation) tuner which allows for the storage of 3 banks of FM presets, FM1/2/3, containing 6 presets each and 3 banks of AM Presets, AM1/AM2/AMA, containing 6 presets each. The AMA bank gives the 6 strongest AM stations stored by an AM autostore. Pre-set stations are stored in the IAM and TS memory. The radio tuner also incorporates the following radio functions:

- HD Radio™ (NAS only)
- AM Auto Store (AST)
- FM station list
- Presets
- Traffic announcements (TA) – Europe only
- RDS (radio data system) / (Radio Broadcast Data System (RBDS) in NAS markets) functions:
 - Station name
 - Radio text
 - PTY
 - Traffic Announcements (TA) (Not NAS)
 - AF switching (Not NAS)
 - REG to lock to regional/local broadcast (Not NAS)
- Seek station
- Tune up/down.

When the optional Digital Radio Control Module (DRCM) is specified, the audio system is fitted with a separate digital radio tuner. This allows the reception of digital broadcast stations which can be received via the FM2/digital radio band III antenna located in the rear spoiler or via the

digital radio band L antenna in the roof pod.

In NAS markets, Satellite Digital Audio Radio Service (SDARS) can be specified; the audio system is fitted with a separate Satellite Radio Control Module (SRCM). This allows for the reception of the satellite digital broadcast stations which are received via a satellite radio antenna located in the roof pod.

All radio antennas are routed into the IAM. The IAM supplies a power output to the AM/FM antenna amplifier.

The IAM incorporates a power management function. Should the vehicle battery level drop below a predetermined level, the unit will limit its functionality. The IAM receives MOST signals from the TS. The TS receives MS (medium speed) CAN comfort bus signals from other vehicle systems which it uses to determine the wake up/shut down process for the modules on the MOST ring.

Calibration of the IAM using approved Land Rover diagnostic equipment enables updates to be downloaded as new technology becomes available or any fault concerns require software updates. If the IAM is replaced it must be configured as a new module using approved Land Rover diagnostic equipment.

CD PLAYER (ALL IAM's)/DVD PLAYER (IAM2 AND 3 ONLY)

The CD player has the capability to play MP3 files. The MP3 discs follow a format of folders and files within the folder. It is also possible to place all the files in the root directory on the virtual CD.

The random and repeat features follow the normal CD random and repeat feature functions.

When a CD error occurs, the IAM will alert the user by showing a message related to the error in the TS. This will be displayed while the CD audio mode is selected until the error is corrected. The CD related error does not affect other areas of the IAM and a different audio source can be selected.

ERROR MESSAGE IN IAM DISPLAY	CAUSE
Mechanism error	Mechanical Error, CD stuck, Servo related error, etc.
Disc error	Invalid Disc, CDROM (read only memory) inserted, disc inserted upside-down, etc.

The IAM3 has the ability to load audio files and 'rip' the music onto the internal hard drive, a 10GB partition is reserved to store music. It is possible to store up to 10 uncompressed albums onto the hard drive. Only CDDA files can be loaded into the virtual changer.

File compatibility for the single slot CD mechanism includes:

- CD audio
- MP3 – (MPEG Layer III)
- WMA – (Microsoft Windows Media Audio)
- WAV – (waveform)
- AAC – (Advanced Audio Coding. Apple iTunes - only through iPod interface).

NOTES:

- The CD player may take a longer time to load an MP3 disc, if there are more tracks than on a normal CD. To minimise loading time, a rigid folder structure is recommended.
- In the event of customer complaints relating to audio quality, file compression should be taken into consideration during diagnosis.

INTEGRATED CONTROL PANEL (ICP)

The ICP is located in the center of the instrument panel, below the TS. The panel is secured with four screws to the instrument panel structure.

The ICP is a plastic molding which is shared with the climate control system.

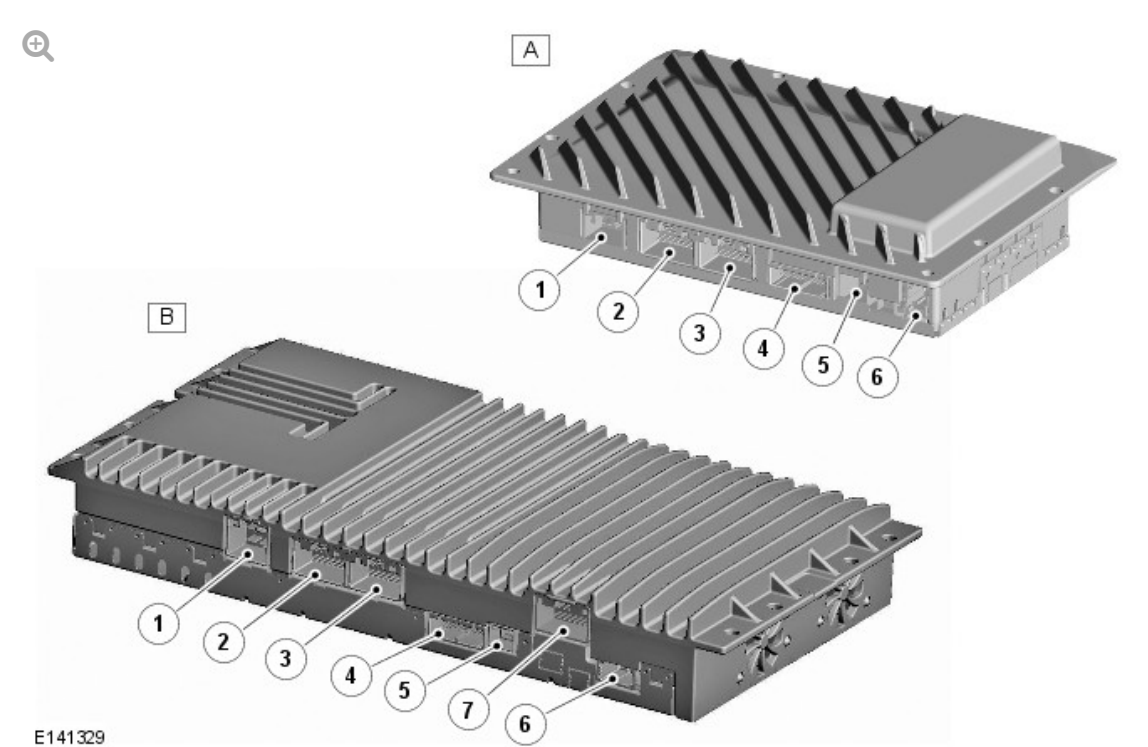
The ICP contains 2 audio controls. A rotary control switch on the left side controls the audio volume by rotation of the control. The audio system can be turned on and off by pressing the control. A switch on the right side ejects a CD/DVD from the IAM.

The ICP has a single connector which provides power and ground, panel illumination and medium speed CAN bus connections for both the ICP and the ATCM (automatic temperature control module) controls

The ICP also contains switches and controls for hazard warning indicators, heated seats and climate control systems. For additional information, refer to:

[Exterior Lighting](#) (417-01 Exterior Lighting, Description and Operation),
Seats (501-10A, Description and Operation),
[Control Components](#) (412-01A Climate Control, Description and Operation).

AUDIO AMPLIFIER MODULE



ITEM	DESCRIPTION
A	Meridian System and Meridian Surround System audio amplifier module

B	Meridian Signature Reference System audio amplifier module
1	Connector - Power from RJB, ground
2	Connector - Speakers
3	Connector - Speakers
4	Connector - Power from RJB, microphones input
5	Connector - headphone transmitter
6	Connector - MOST
7	Connector - Speaker (Meridian Signature Reference system only)

The audio amplifier module is located below the left front seat. The audio amplifier module is attached to 2 brackets which are secured to 4 studs the vehicle floor pan and secured with flanged nuts.

The system has 3 audio amplifier module options depending on the audio system fitted:

- 384W - Meridian system
- 880W - Meridian Surround system
- 1700W - Meridian Signature Reference system.

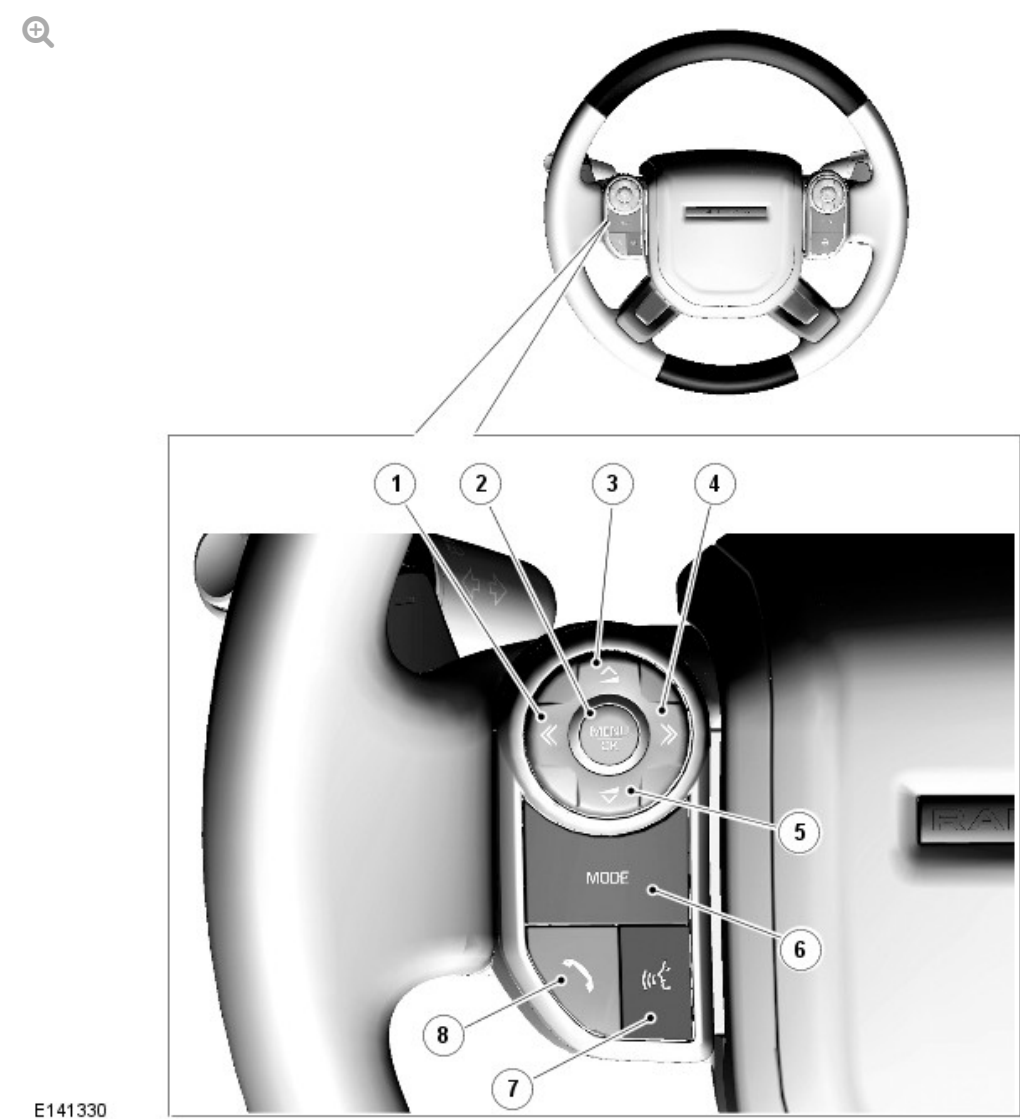
The audio amplifier module is connected to the other audio system components on the MOST ring. The amplifier module processes incoming audio signals from other system modules on the MOST ring. Hardwired connectors provide outputs for the vehicle speakers, output to the headphone transmitter and power and ground. Power supplies are received from the RJB (rear junction box).

In conjunction with the 29 speakers, the AUD3-24 channel Meridian Signature Reference system audio amplifier module produces 1700W output and features Meridian Trifield, Meridian Trifield 3D, Dolby Pro Logic IIx, Dolby Pro Logic IIz, and DTS Neo:X surround sound.

Dynamic Equalization (EQ) - Meridian Signature Reference System Only

Four dynamic EQ microphones, located in the headliner, are used for Dynamic EQ. The system uses the four microphones to measure noise within the vehicle (for example, wind or road noise). This information is processed by the audio amplifier module, which calculates which frequencies being played need to be compensated due to the noise. The audio amplifier module then adjusts the volume, bass and treble outputs to counter the noise inside the vehicle.

LEFT STEERING WHEEL SWITCHPACK (AUDIO)



ITEM	DESCRIPTION
1	Seek, previous radio preset, previous track, scroll down telephone list
2	Menu/OK button (instrument cluster menu control)
3	Volume increase

4	Seek, next radio preset, next track, scroll up telephone list
5	Volume decrease
6	Mode button
7	Voice button
8	Telephone answer/end call button

The left steering wheel switchpack is located on the left side of the steering wheel.

The switches are a resistive ladder type which apply a different voltage to the TS in response to different switches being pressed.

The left steering wheel switchpack controls the following functions:

- MODE - Press repeatedly to scroll through all audio/video sources.
- » Short press up:
 - to select the next radio preset
 - to select the next track on chosen audio source
 - when telephone is in use, press to scroll up lists of calls or phonebook entries.
- «Short press down:
 - to select the previous radio preset
 - to select the previous track or start of current track on chosen audio source
 - when telephone is in use, press to scroll down lists of calls or phonebook entries.
- With radio manual seek mode activated, further short presses will change the frequency in single increments. A further long press will scan forwards through the current waveband until the button is released
- » Long press up:
 - to auto seek up the frequency to the next radio station.

- « Long press down:
 - to auto seek down the frequency to the next radio station.
-
- Volume increase for any audio source
- Volume decrease for any audio source.

The MENU/OK button is for use with the instrument cluster menu.

For additional information, refer to: [Instrument Cluster](#) (413-01 Instrument Cluster, Description and Operation).

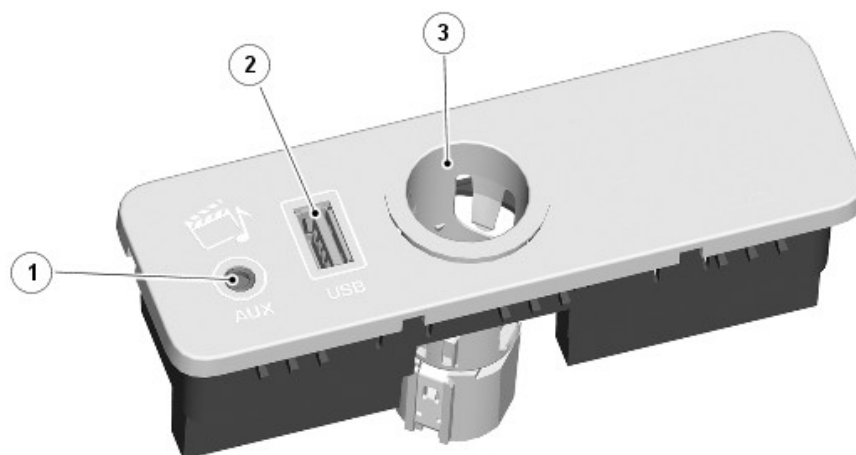
For further details on the telephone button refer to Cellular Phone.

For additional information, refer to: [Cellular Phone](#) (415-01 Information and Entertainment System, Description and Operation).

For further details on the voice button refer to Voice Control.

For additional information, refer to: [Voice Control](#) (415-01 Information and Entertainment System, Description and Operation).

PORTABLE AUDIO INTERFACE PANEL



E142557

ITEM	DESCRIPTION
1	3.5mm jack plug - MP3 connection
2	Universal Serial Bus (USB) connector
3	12V accessory socket

The portable audio interface panel is located in the floor console below the stowage compartment lid and is fitted to all audio systems. The interface is a media hub between a portable input device and the IAM. The interface contains one USB port and a 3.5mm jack plug connector.

Devices that can be connected to the portable audio interface panel include:

- USB mass storage devices (for example a memory stick). Devices must use FAT or FAT32 file format.
- iPod® (iPod Classic, iPod touch, iPhone and iPod Nano are supported - full functionality for older devices cannot be guaranteed). iPod Shuffle functionality cannot be guaranteed.
- Auxiliary device (personal audio, MP3 players).
- Devices with Bluetooth® connectivity. Devices must support A2DP and AVRCP Bluetooth® protocols).

When connecting a portable storage device, select AUX on the TS to select that device input. Depending on the device connected, many of the audio controls on the TS and steering wheel switches can be used.

NOTE:

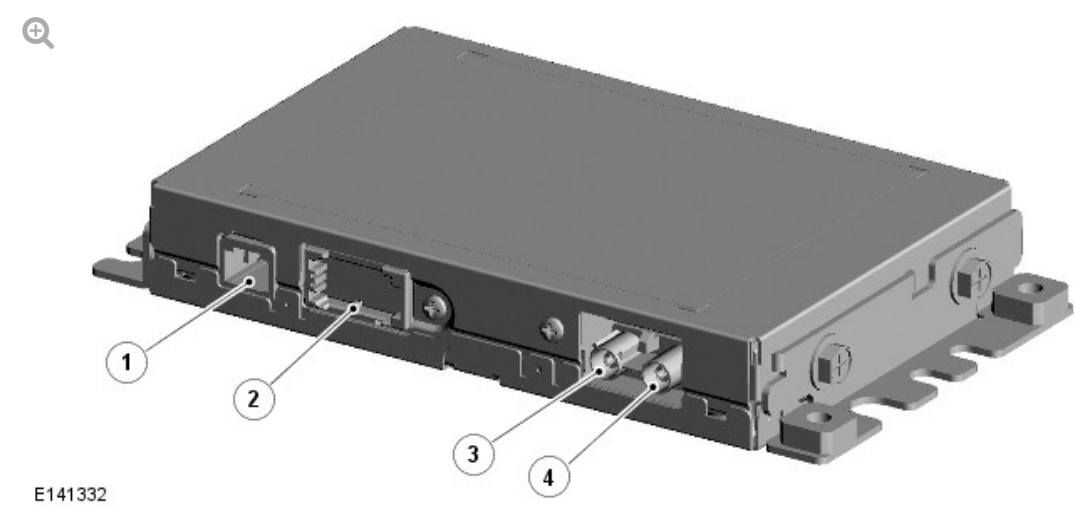
It is recommended to disconnect an iPod when leaving the vehicle. Failure to do so may result in the iPod battery discharging.

To maximise playback quality, it is recommended to use lossless compression for any media files on iPod. Failing this, it is recommended that compressed files utilise a minimum bitrate of 192 kb/s (a higher bitrate is strongly recommended for increased quality playback).

For media files delivered via USB or MP3 players, the highest compression rate supported is 320 kbits per second (kbps). If anything less than 128 kbps is used, Digital Signal Processing (DSP) functionality may be lost.

Some MP3 players may have a file system that is not supported by the IAM. To use an MP3 player, it must be set to USB Removable Device or Mass Storage Device mode. Only music that has been added to the device in this mode can be played.

DIGITAL RADIO CONTROL MODULE



ITEM	DESCRIPTION
1	Connector - power and ground
2	Connector - MOST
3	Connector - L Band antenna input
4	Connector - Band III antenna input

The digital radio control module is located on the floor of the luggage compartment, on the left side. The module has mounting brackets that are located on studs and secured with flanged nuts.

Digital Radio is designed to provide reliable, multi-service broadcasting for reception by mobile, portable and fixed tuners.

Digital radio provides a clear signal with minimal interference, hiss or fading. After a channel (or station) has been tuned and stored, it does not need retuning.

NOTE:

Radio signals travel in a straight line so large obstacles, such as tall buildings, can shield the vehicle from the signal causing temporary loss of reception (known as dead spots). The digital radio control module selects the strongest radio transmitter, minimizing signal loss.

Digital radio is transmitted from regional terrestrial transmitters. Local digital radio channels are not available outside the range of a transmitter. To receive new local channels during vehicle movement around a country, the auto-tune function is used to build new channel lists.

NOTE:

When the vehicle digital radio is first used the system will not receive any digital stations until the auto-tune function has been completed.

Digital radio channels are organized into groups called ensembles (also known as multiplexes). Some individual channels may also provide a number of subchannels. For example, if several sports events are being held simultaneously, the channel may temporarily choose to broadcast each different event on a separate subchannel.

Digital radio is broadcast in 3 formats on a compatible network structure. The digital radio control module is capable of processing all digital formats and is therefore not market specific. The 3 formats are:

- Digital Audio Broadcasting (DAB) - MP3 coding
- Digital Audio Broadcasting + (DAB+) - MP3 and MP4 coding
- Digital Multimedia Broadcasting (DMB) - MP3 and MP 4 coding with video/multimedia.

Digital radio is currently broadcast across Europe, Canada, Australia and

parts of Asia. System transmission is via a terrestrial network, on two separate broadcasting bands:

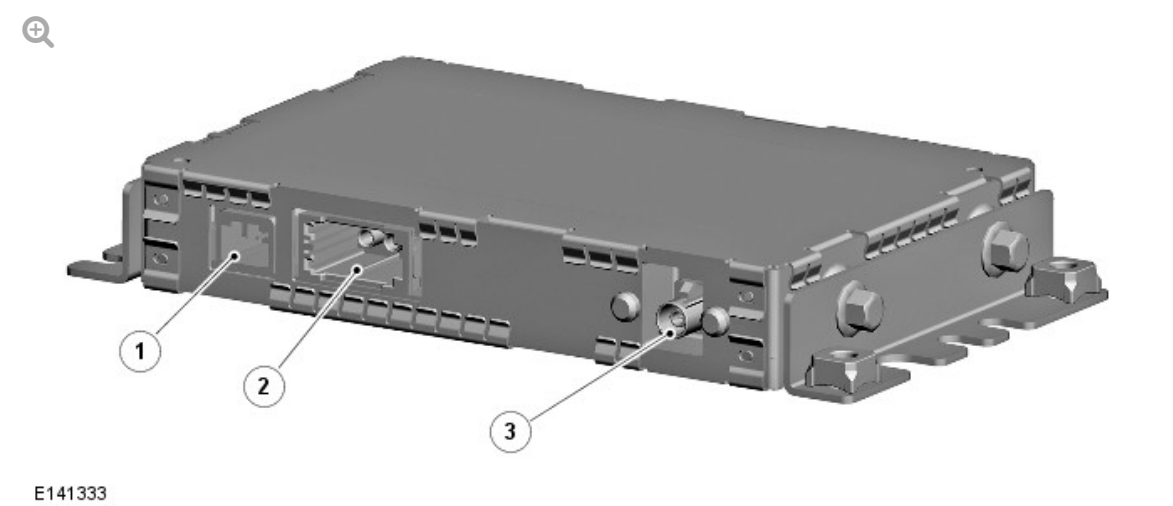
- Band-L - Received from roof pod antenna
- Band III - Received from spoiler mounted antenna.

Operation of the digital radio system is the same as the radio operation with selections made through the TS to access and navigate the system functions.

The digital radio control module is a dedicated tuner which is controlled by the TS on the MOST ring. The tuner processes the signals from the digital radio antennas. Information is transmitted on the MOST ring and processed by the IAM. The processed information is sent out to the audio amplifier module and broadcast through the speaker system.

No configuration procedure is required if the digital radio control module is replaced. Calibration of the module using Land Rover approved diagnostic equipment enables updates to be downloaded as new technology becomes available or any fault concerns require software updates.

SATELLITE RADIO CONTROL MODULE (NAS ONLY)



E141333	
ITEM	DESCRIPTION
1	Connector - power and ground

2	Connector - MOST
3	Connector - satellite radio antenna input

The satellite radio control module is located on the floor of the luggage compartment, on the left side. The module has mounting brackets that are located on studs and secured with flanged nuts.

The Satellite radio control module is a dedicated tuner which is controlled by the IAM on the MOST ring. The tuner filters the signals from the satellite radio antenna located in the roof pod or sigma pod. Information is transmitted on the MOST ring and processed by the IAM. The processed information is sent out to the audio amplifier module and broadcast through the speaker system.

The digital radio format adopted for NAS vehicles is satellite radio operating in the S-Band frequency range (2.3 GHz). Satellite Digital Audio Radio Service (SDARS) service provider 'Sirius®' transmit a signal from their up-link facility (which is the original point of transmission of data, voice or other information through an antenna system) to a satellite where the signal is then down linked to both the terrestrial repeater network and the individual satellite radio system in the vehicle. The satellite radio control module switches between the satellite signal and the repeater network signal depending on the strength of the signal at any given time.

The Sirius® satellite system comprises:

- Satellites
- Ground repeaters
- Up-link ground stations
- Radio receiver systems

The Sirius® satellite radio system uses three satellites on an inclined elliptical orbit. This ensures that each satellite spends approximately 16 hours a day over the continent of North America, with at least one satellite over the country at any one time.

The satellites beam their signals down to the ground where the signal is picked up by receivers or is transmitted to repeater stations to cover built up areas where the signal is obscured. The satellite service comprises over 100 channels of digital entertainment which is provided by subscription requiring a monthly payment. In order to obtain a subscription, the satellite radio control module ID number will need to be retrieved from the module. This is achieved as follows:

- When the vehicle first receives live signal within the US or Canada, the TS will display ch184 as the tuned channel (labelled as Preview). The radio text will display Call 1888-539-Sirius. Alternatively, when the radio is subscribed, unsubscribe channels shall also display this information in the radio text
- The ESN/SID can always be found the following way, regardless of the subscription status of the radio. Press **Settings** on the TS, then select **Sat Info** icon in the settings view. This will display the ESN/SID and the phone number. If a telephone is paired to the Bluetooth® system, the user can select the green phone button in the Sat info view to call the call center.

If no subscription has been taken, the TS will display the Sirius® telephone number. To subscribe to Sirius® use the displayed phone this number. The user will need payment details, the Sirius® ID number and details of the required package.

The satellite radio function is accessed by pressing the **SAT Radio** soft key. This allows the user access to another bank of possible favourites, with the option to store 6 favourites on each of SAT 1, SAT 2 and SAT 3.

Operation of the satellite radio system is the same as the radio operations with selections made through the TS to access and navigate the system functions.

No configuration procedure is required if the satellite radio control module is replaced. Calibration of the satellite radio control module using approved Land Rover diagnostic equipment enables updates to be downloaded as

new technology becomes available or any fault concerns require software updates.

HD RADIO™ (NAS ONLY)

HD Radio™ technology is a free digital radio format broadcast available in the NAS market and may become available for other markets in the future.

HD Radio™ digital broadcasts are transmitted alongside the analogue AM and FM signals by stations broadcasting HD Radio™ signals. HD Radio™ technology has the ability to deliver improved sound quality and content to the listener, blending between analogue and a digital audio streams of the primary station and delivering extra multicast stations on the same frequency.

The HD Radio™ receiver is integrated into the AM / FM tuner on NAS variants of the IAM.

ANTENNAS

AM, FM, digital radio, satellite radio, TV (television) and other system antennas are located in the rear spoiler, roof pod and side windows. Refer to the Antennas section.

For additional information, refer to: [Antenna](#) (415-01 Information and Entertainment System, Description and Operation).

SPEAKERS

The number, location and specification of the speakers in the vehicle differs depending on the level of audio system fitted.

For further details of the speaker systems, refer to the Speakers section.

For additional information, refer to: [Speakers](#) (415-01 Information and Entertainment System, Description and Operation).

ADVANCED VOICE CONTROL SYSTEM

The advanced voice system provides the driver with the option of voice control for a range of supported functions. The following systems include Voice functionality:

- Radio, satellite radio and Digital Radio (if fitted)
- CD player
- Hard disk drive/virtual CD player
- USB and auxiliary connections
- Navigation system
- Bluetooth® Telephone system
- Vehicle notepad.

For further details of the advanced voice control system refer to the Voice Control section.

For additional information, refer to: [Voice Control](#) (415-01 Information and Entertainment System, Description and Operation).

OPERATION

MEDIA ORIENTED SYSTEMS TRANSPORT (MOST)

The components of the audio/infotainment systems are all connected on the Media Oriented Systems Transport (MOST) ring. The MOST ring is a fiber optic communications bus for multimedia applications. Audio and control information is passed around the MOST ring and can be picked up by any of the systems units. For example, radio station tuning/selection input by the vehicle user into the Touch Screen (TS) is sent along the MOST ring and collected by the Integrated Audio Module (IAM) which then selects the requested radio station.

MOST technology uses a plastic optical fiber which forms a network connecting the audio and multimedia system components. Each component in the ring is connected to the plastic optical fiber through a device known as a Fiber Optical Transceiver (FOT). Each FOT has two optical connections; one connection is sensitive to light and is the input, the second connection forms the light source and is the output. The system operates by connecting the output from one FOT to the input of another FOT.

The light signals are sent in one direction only and are formed in the following way:

- Electrical signals are converted into an electrical current
- The current then drives an LED in the FOT to produce a high intensity red light
- The LED transmits the light through a fiber optic cable
- A photo diode in the FOT at the opposite end of the fiber optic cable detects the light.

The following components may be connected to the MOST ring dependant on the vehicle equipment level:

- IAM
- TS
- Digital Radio control module (if fitted)
- Satellite radio control module (NAS only)
- Audio amplifier module
- Navigation computer (Japan only)
- Rear Seat Entertainment (RSE) control module
- TV tuner (if fitted)
- Instrument cluster.

MOST is a synchronous network. A timing master supplies the clock information and all other devices on the network synchronize their operation to this clock. The timing master for the MOST network on this vehicle is the TS. This TS controls and manages the MOST ring and the system components. An Optical Bus tester is used in conjunction with approved Land Rover diagnostic equipment to diagnose the MOST system.

The Optical Bus tester emits a visible, high intensity red light which can be connected into the ring at any point to test the ring integrity. Disconnecting

a MOST connector will reveal if the high intensity red light is visible. If a break occurs in the MOST ring, fault codes are stored in the TS which can be retrieved using approved Land Rover diagnostic equipment.

The MOST connection to the instrument cluster allows audio, navigation, telephone and voice control information to be displayed in the instrument cluster TFT display.

For additional information, refer to: [Instrument Cluster](#) (413-01 Instrument Cluster, Description and Operation).

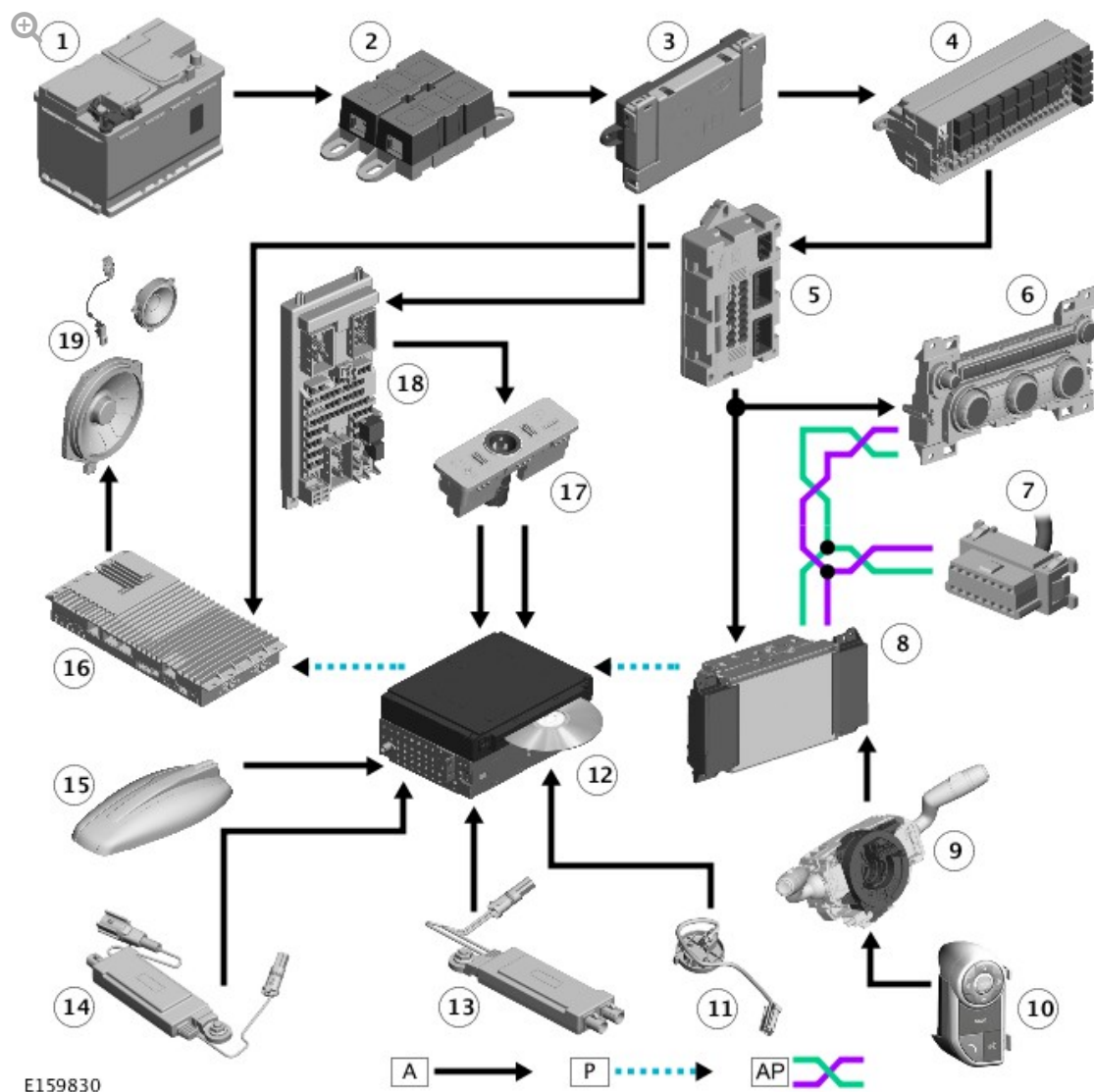
Audio System Gateway Functions

The TS is the gateway between MS (medium speed) CAN comfort bus and the MOST systems. A typical example of information transfer is vehicle speed information from the ABS (anti-lock brake system) control module used to control the dynamic equalization function (where fitted). The vehicle speed information from the ABS control module is sent on the HS (high speed) CAN chassis bus and collected by the instrument cluster gateway.

The signal is passed to the MS CAN comfort bus to the TS. The TS passes the information on the MOST to audio amplifier module which also uses the inputs from the microphones to calculate the volume adjustment required.

The calculations are performed by the audio amplifier module which adjusts the output to the speaker system.

CONTROL DIAGRAM - AUDIO SYSTEM

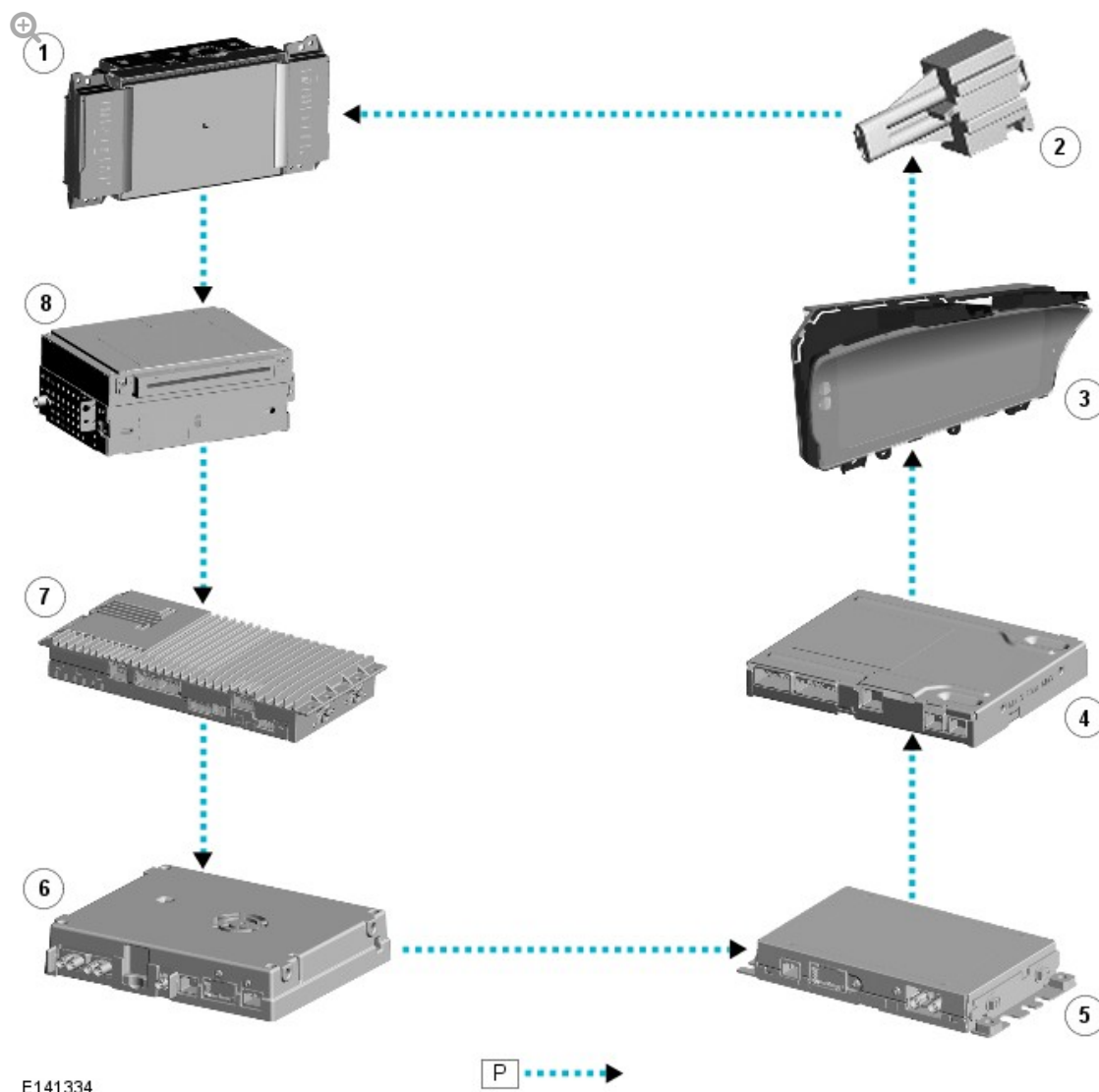


A = HARDWIRED; P = MOST (MEDIA ORIENTED SYSTEM TRANSFER);
AP = MS (MEDIUM SPEED) CAN COMFORT BUS.

ITEM	DESCRIPTION
1	Battery
2	2 (BJB2)
3	Battery Junction Box (BJB)
4	Rear Junction Box (RJB)
5	Quiescent Current Control Module (QCCM)
6	Integrated Control Panel (ICP)
7	Diagnostic socket
8	Touch Screen (TS)

9	Clockspring
10	Left steering wheel audio switches
11	Microphone (Voice Control)
12	Integrated Audio Module (IAM)
13	FM2 / Digital Radio / TMC / VICS (Japan only) antenna amplifier
14	AM / FM antenna amplifier
15	Roof pod (satellite radio antenna NAS only) / (GPS antenna Ref only)
16	Audio Amplifier Module (AAM)
17	Portable audio interface panel
18	Central Junction Box (CJB)
19	Vehicle speaker system

CONTROL DIAGRAM - MOST



P = MOST;

ITEM	DESCRIPTION
1	Touch Screen (TS)
2	MOST diagnostic connector
3	Instrument cluster
4	Rear Seat Entertainment (RSE) control module
5	Digital radio control module (ROW) / Satellite radio control module (NAS)
6	TV control module
7	Audio amplifier module
8	Integrated Audio Module (IAM)

